■# PAPER_312: NGC 6302 Central Star UV Radiation Pressure — UQFF Photoionization Gravitational Signature

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■# PAPER_312: NGC 6302 Central Star UV Radiation Pressure — UQFF Photoionization Gravitational Signature

Subtitle: FIRST UQFF Hot-WD UV Radiation Parameter — $\eta rad = 1.913\times10^{20}$; arad = 5.672×10■ m/s^2

Author: Daniel T. Murphy **Session:** 89 | **Date:** March 17, 2026 **Module:** `NGC6302_UQFF_MODULE.cpp` (31st C++ UQFF module) **WOLFRAM_TERM:** `NGC6302_UV_RADIATION` **UQFF First:** FIRST UQFF explicit UV-bright white dwarf (T_eff = 200,000 K) photon-pressure gravitational signature

<!-- UQFF constants: κ = 5.0e-4 day^{-1}, [SSq] = 0.57, M_UQFF = 1.43e1 TeV -->

Abstract

This paper presents UQFF derivations and numerical results for: PAPER_312: NGC 6302 Central Star UV Radiation Pressure — UQFF Photoionization Gravitational Signature. Calibration constants: `\kappa` = 0.0005/day, [SSq] = 0.57. Results validated against observational data and prior UQFF whitepaper series.

1. System Parameters

Parameter	Value	Notes
T_{eff}star	2.0×10■ K	Central WD effective temperature
L_star	1.914×10^{30} W	= 5000 L_sun (Zanstra temperature analysis)
r	9.46×10^{1}■ m	PN half-lobe radius
ρ_fluid	1.0×10^{-20} kg/m^3	Ionized lobe gas density
c	3.0×10■ m/s	Speed of light

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2. Unique Physics

2.1 Central Star Luminosity

NGC 6302's central star is one of the hottest white dwarfs known, with T_eff ≈ 200,000 K (Szyszka et al. 2009, ApJL). The Zanstra hydrogen luminosity gives:

```
L_{star} = 5000\ L_\odot = 5000 \times 3.828 \times 10^{26}\ \text{W} = 1.914 \times 10^{30}\ \text{W}
```

2.2 Radiation Pressure at Lobe Radius

The photon radiation pressure at lobe radius r for an isotropic point source:

```
P_{rad} = \frac{L_{star}}{4\pi r^2 c}
```

```
4\pi r^2 = 4\pi \times (9.46 \times 10^{15})^2 = 4\pi \times 8.949 \times 10^{31} = 1.125
\times 10^{33}\ \text{m}^2
```

```
P_{rad} = \frac{1.914 \times 10^{30}}{1.125 \times 10^{33} \times 3.0 \times 10^8} =
\frac{1.914 \times 10^{30}}{3.375 \times 10^{41}} = \mathbf{5.672 \times 10^{-12}\
\text{Pa}}
```

2.3 Radiation-Pressure Acceleration

```
a_{rad} = \frac{P_{rad}}{\rho_{fluid}} = \frac{5.672 \times 10^{-12}}{1.0 \times
10^{-20}} = \mathbf{5.672 \times 10^8\ \text{m/s}^2}
```

2.4 Radiation-to-Gravity Dominance Ratio

```
\eta_{rad} \equiv \frac{a_{rad}}{g_{base}} = \frac{5.672 \times 10^8}{2.967 \times
10^{-12}} = \mathbf{1.913 \times 10^{20}}
```

The UV radiation pressure exceeds gravitational force by $\mathbf{1.913\times10^{20}}$ — twenty orders of magnitude. This establishes that photoionization-driven radiation pressure is the primary mechanism for lobe acceleration in bipolar PNe with ultra-hot central stars.

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3. UQFF Pipeline Term

In the full UQFF 2.0 pipeline, the UV radiation acceleration enters as an independent additive term:

```
g_{NGC6302}^{(rad)} = \frac{L_{star}}{4\pi r^2\ c\ \rho_{fluid}}
```

This term dominates over the wind shock term (PAPER_311: `a_{wind} \sim 10^{-6}` m/s^2) by a further **14 orders of magnitude**, placing radiation pressure at the apex of the NGC 6302 UQFF force hierarchy.

Force Hierarchy (descending):

`a_{rad} = 5.672 \times 10^8` m/s^2 — UV radiation pressure (PAPER_312, **DOMINANT**)

`a_{wind} = 2.114 \times 10^{-6}` m/s^2 — wind shock (PAPER_311)

`g_{base} = 2.967 \times 10^{-12}` m/s^2 — gravitational binding

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4. Astrophysical Context

The UV radiation from NGC 6302's central star (T*eff = 200,000 K) photoionizes the surrounding gas, producing the characteristic bipolar emission nebula observed in [O III], H-alpha, and UV by HST/WFC3. The radiation pressure parameter* ηrad = 1.913×10^{20} confirms that the nebular gas is radiation-pressure dominated at all scales up to the lobe boundary.

The UQFF formulation explicitly identifies this as a distinct gravitational-equivalent acceleration channel, separable from wind shock effects and magnetic confinement — providing the first three-component force budget for a bipolar PN within the UQFF framework.

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5. Key Results

Quantity	Value	Unit
L*star (5000 L*sun)	1.914×10^{30}	W
P_rad at r=1 ly	5.672×10^{-12}	Pa
a_rad	**5.672×10■**	m/s^2
η_rad	**1.913×10^{20}**	dimensionless
a*rad / a*wind	2.684×10^{1}■	dimensionless

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6. Classification

UQFF First: FIRST UQFF explicit hot-WD UV radiation parameter (T_eff = 200,000 K class)

Scale: Stellar (PN lobe scale, ~1 ly)

Physics category: Photoionization / UV radiation pressure / force hierarchy

Cross-references: PAPER*311 (wind shock), PAPER*313 (torus magnetic confinement)
